Do Now · retrieval

Software testing: finding and correcting faults

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Define a cache miss.

Q2

When is A AND B true?

Q3

Give a boundary-invalid input just below 1.

Q4

Does 61 satisfy `seconds>=1 || seconds<=60`?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Read the stated rule — Because the permitted ages are inclusive, both age 12 and age 18 must be accepted.

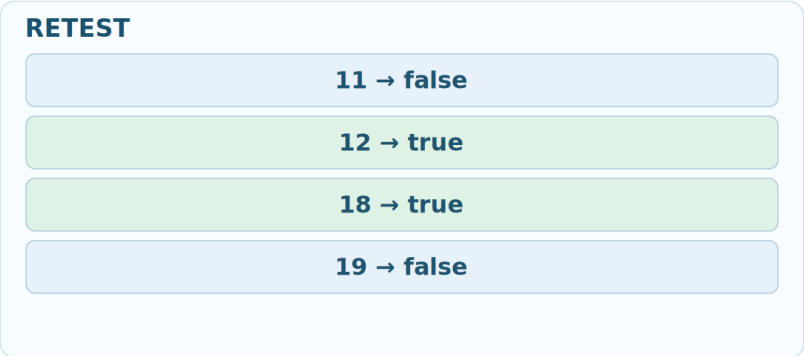
Core idea 2

Locate the Boolean error — The strict comparison operators > and < exclude both valid boundaries.

Core idea 3

Repair and test — Change to >= and <=; retest the boundaries plus values just outside the range.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

logic error — Runs but produces the wrong result.

inclusive boundary — Endpoint allowed by a range.

validation — Check input meets constraints.

debugging — Find and correct a fault.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

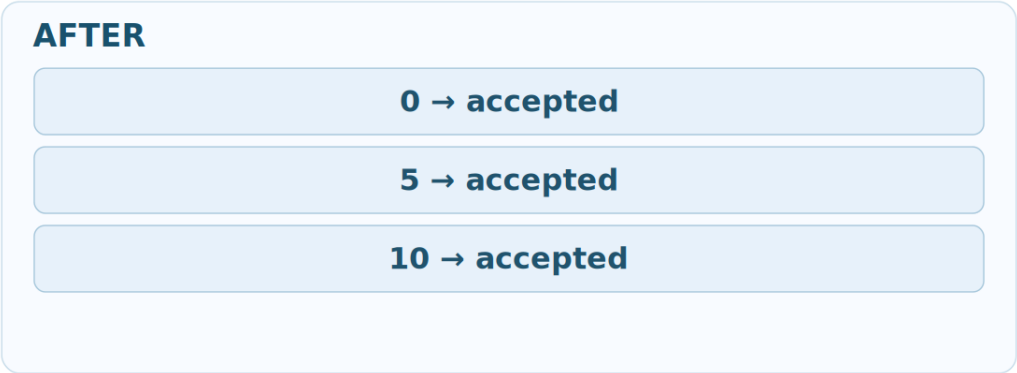
Problem

A score validator intended for 0–10 uses score>0 && score<10. Fix it.

Follow the worked method

1. Predict first — The valid endpoints were incorrectly rejected.
2. Model and check — Change the operators to >= and <=, and test −1 and 11 as invalid values.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
OR accepts 61 because only one side needs to be true.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Correct x>5 && x<15 for values 5–15 inclusive, and state whether 5 and 15 are valid.

Question 2

Explain why `age>=12 || age<=18` accepts 100, correct it, and give outputs for 11,12,18,19.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Outline a debugging and regression-testing process after correcting a validator.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

For a bounded range, OR usually replaces AND without changing the outcome.

Use key vocabulary from memory

1. Explain logic error in your own words.

2. Explain inclusive boundary in your own words.

3. Explain validation in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.